SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION **COURSE CODE: DSA0204**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- **Viola-Jones: fast but less accurate (78%)**
- **YOLO: moderate speed, good accuracy (92%)**
- **Deep Learning: high accuracy (95%) but slow**

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

1. Real-Time System Design Trade-off

A surveillance system must detect objects quickly while maintaining acceptable accuracy. The three techniques have different advantages.

Technique	**Speed**	**Accuracy**	**Computational Cost**	**Real-Time Suitability**
Viola-Jones	Very High	78%	Low	Excellent
YOLO	Moderate/High	92%	Medium	**Excellent**
Deep Learning	Low	95%	High	Limited

Viola-Jones

Viola-Jones is computationally lightweight and can process images very quickly. It is useful when the system has very strict latency requirements and limited hardware resources.

However, its accuracy is only **78%**. In a surveillance system, this can lead to missed objects and false detections. Therefore, although it is fast, its lower accuracy can reduce system reliability.

YOLO

YOLO provides **92% accuracy** with moderate processing speed. It can detect objects in a single pass through the image, making it suitable for real-time applications.

It provides a much better accuracy than Viola-Jones while still maintaining reasonable speed.

Deep Learning

Deep Learning achieves the highest accuracy of **95%**, but it requires more computational resources and has slower processing. It may require GPUs or specialized hardware for real-time deployment.

Evaluation

The difference between YOLO and Deep Learning is:

95% − 92% = 3% accuracy improvement

This small improvement may not justify the large increase in computational cost and latency when the system has strict response-time requirements.

Final Decision

YOLO should be selected.

It provides the best balance between:

- High accuracy
- Fast detection

- Moderate computational requirements
- Real-time operation
- Good performance in practical surveillance environments

Conclusion: If latency is extremely strict and hardware is very limited, Viola-Jones can be considered. However, when **acceptable accuracy and real-time performance must both be maintained, YOLO is the preferred solution.**

2. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method	Normal	Occlusion	Low Light
Viola-Jones	**80%**	**60%**	**55%**
YOLO	**90%**	**75%**	**70%**
Deep Learning	**95%**	**85%**	**80%**

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

The system must work under normal conditions, object occlusion, and low-light conditions.

Method	Normal	Occlusion	Low Light	Average Accuracy
Viola-Jones	80%	60%	55%	**65%**
YOLO	90%	75%	70%	**78.33%**
Deep Learning	95%	85%	80%	**86.67%**

Step-by-step comparison

Viola-Jones

- Normal: 80%
- Occlusion: 60%
- Low light: 55%

It performs reasonably under normal conditions but its performance decreases significantly when the env

ironment becomes difficult.

The drop from normal to low light is:

80 − 55 = 25 percentage points

Therefore, it is less robust.

YOLO

YOLO performs better:

- Normal: 90%
- Occlusion: 75%
- Low light: 70%

Its performance decreases under difficult conditions, but it still maintains relatively good detection.

Deep Learning

Deep Learning provides:

- Normal: 95%
- Occlusion: 85%
- Low light: 80%

It has the highest accuracy in every condition.

Its decrease from normal to low light is:

$95 - 80 = 15$ percentage points

This is smaller than the decrease for Viola-Jones, indicating better robustness.

Final Decision

Deep Learning is the best technique for robust detection.

It is particularly suitable for:

- Outdoor surveillance
- Night-time monitoring
- Crowded environments
- Partially hidden objects
- Complex backgrounds
- Changing illumination

Trade-off

The main disadvantage is computational cost.

Therefore:

Maximum robustness required → Deep Learning

Balanced speed and robustness → YOLO

3. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- **Method A: stable vectors but misses small motions**
- **Method B: detects fine motion but noisy vectors**

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

Optical flow estimates the movement of objects between consecutive frames.

The two methods have opposite characteristics.

Factor	Method A	Method B
Vector stability	**High**	Low
Noise sensitivity	Low	**High**
Small motion detection	Poor	**Excellent**
Tracking reliability	**High**	Moderate
Fine motion detection	Poor	**Excellent**

Method A – Stable but misses small motion

Method A generates stable motion vectors. This means the estimated movement is consistent between frames.

Advantages

- Reliable motion estimation
- Less affected by noise
- Smooth tracking
- Suitable for large object movement

Disadvantage

It may fail to detect small movements.

For example, if a person's hand moves slightly while the body remains stationary, Method A may not capture the movement effectively.

Method B – Sensitive but noisy

Method B can identify very small movements.

Advantages

- Detects fine movements
- Good for subtle motion
- Useful for detailed motion analysis

Disadvantages

- Sensitive to image noise
- Motion vectors may fluctuate
- Can create false motion
- Requires filtering or smoothing

Final Decision

For **general dynamic-scene tracking**, Method A is generally safer because stable tracking is important.

For applications where detecting **small movements** is the primary requirement, Method B is better.

A good practical solution is:

Method B + noise filtering/smoothing

This provides the ability to detect fine movements while reducing unwanted noise.

4. Background Modeling Strategy (GMM)

A GMM-based system shows:

- **High false positives in dynamic background**
- **Missed detections in static background**

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

A Gaussian Mixture Model (GMM) is commonly used for background subtraction. It models the background using multiple Gaussian distributions and identifies pixels that differ from the background as foreground.

However, the given system has two problems.

Scenario 1: Dynamic Background

The system produces high false positives.

For example:

- Moving tree branches
- Water movement
- Rain
- Camera vibration
- Moving shadows

These can be incorrectly classified as foreground objects.

Result

False Positive ↑

This reduces detection accuracy.

Scenario 2: Static Background

The system produces **missed detections**.

This can happen when an object remains stationary for a long time. The background model may gradually adapt and consider the object part of the background.

For example, if a person enters a scene and stands still for several minutes, GMM may eventually treat that person as part of the background.

Result

False Negative ↑

Is GMM Alone Sufficient?

No.

GMM is useful for basic foreground detection, but it has limitations in complex environments.

Recommended Improvements

1. Adaptive Background Updating

The background model should continuously adapt to environmental changes.

2. Shadow Detection

Shadow removal can prevent shadows from being classified as objects.

3. Morphological Operations

Operations such as:

- Opening
- Closing
- Erosion
- Dilation

can remove small noise and fill gaps in detected objects.

4. Optical Flow

Optical flow can provide additional motion information.

A combined system can use:

GMM → Foreground Detection

Optical Flow → Motion Verification

5. YOLO/Deep Learning

An object detector can identify whether the detected foreground region is actually a person, vehicle, animal, etc.

Better Architecture

Video → GMM → Noise/Shadow Removal → Optical Flow → YOLO → Final Detection

This hybrid approach is more reliable than GMM alone.

Final Decision

GMM is useful as a **background subtraction component**, but it should not be the only detection technique in a complex surveillance system.

5. Motion Estimation vs Computational Cost

Two methods:

- **Method A: low error (5%), high computation**
- **Method B: moderate error (10%), low computation**

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

5. Motion Estimation vs Computational Cost

The two methods represent a classic accuracy versus computation trade-off.

Factor	Method A	Method B
Error	5%	10%
Accuracy	95%	90%
Computation	High	Low
Processing speed	Slow	Fast
Real-time suitability	Moderate	Excellent
Offline suitability	Excellent	Good

Method A

Method A has only 5% error, meaning approximately 95% accuracy.

Advantages

- High accuracy
- Better motion estimation
- More precise results
- Suitable when quality is more important than speed

Disadvantage

It requires high computational resources.

Therefore, it may not be suitable for systems where results must be generated immediately.

Method B

Method B has 10% error, corresponding to approximately 90% accuracy.

Although its accuracy is lower, it requires much less computation.

Advantages

- Fast processing
- Low hardware requirements
- Suitable for real-time systems
- Lower energy consumption

Disadvantage

It produces less accurate motion estimates.

Real-Time Application

For real-time applications, Method B is preferred.

Examples:

- Live surveillance
- Real-time traffic monitoring
- Drone tracking
- Video conferencing
- Real-time robotics

In these applications, a small delay can reduce system usefulness.

Therefore:

Low computation + fast response > slightly higher accuracy

Offline Application

For offline applications, Method A is preferred.

Examples:

- Video analysis
- Scientific research
- Forensic analysis
- Post-event surveillance analysis
- High-quality motion estimation

Since the system does not need to respond immediately, the additional computation can be accepted.

Therefore:

Higher accuracy > processing speed

Overall Comparison

Question	Best Choice	Main Reason
1. Real-Time Surveillance	**YOLO**	Best balance of speed and accuracy
2. Environmental Robustness	**Deep Learning**	Highest accuracy across all conditions
3. Optical Flow	**Method A for general tracking**	Stable and reliable vectors
4. Background Modeling	**Hybrid GMM approach**	GMM alone has false positives/negatives
5. Motion Estimation	**Method B for real-time; A for offline**	Choice depends on latency vs accuracy

Code of Conduct:

I M Jeevitha (192424257) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jeevitha

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**

Understanding of Data	**25%**	**Accurately interprets all data patterns, trends, and relationships**	**Interprets data correctly with minor gaps**	**Partial understanding; misses key trends**	**Misinterprets or fails to understand data**
Analysis	**25%**	**Provides deep analysis with strong logical reasoning and justification**	**Provides reasonable analysis with some justification**	**Basic analysis with weak reasoning**	**No meaningful analysis provided**
Application of Concepts	**20%**	**Effectively applies relevant concepts/tools to interpret data accurately**	**Applies concepts with minor errors**	**Limited application; requires guidance**	**Unable to apply concepts to data**
Conclusion	**20%**	**Draws clear, meaningful conclusions with strong insights and implications**	**Provides valid conclusions with moderate insights**	**Conclusions are vague or weak**	**No clear or valid conclusions**
Clarity	**10%**	**Presents interpretation clearly using proper structure, visuals, and terminology**	**Generally clear with minor issues**	**Lacks clarity or proper structure**	**Poor presentation and difficult to understand**